Checkerboard calibration target, composite

DATASHEET Rev A · 2026-10-24

The classic pattern: highest corner localisation accuracy of any planar target when the whole board is in view. Supported by every calibration toolchain. Choose Fine for more corners per image (higher-resolution cameras, shorter distances) or Coarse for low-resolution or long-distance setups.

Ordering

Part number	Size (W × H)	Fine (F): squares · size	Coarse (C): squares · size
CT-CHT-P-200150-{F\	C}	200 × 150 mm	7 × 10 · 18 mm
CT-CHT-P-300200-{F\	C}	300 × 200 mm	9 × 14 · 20 mm
CT-CHT-P-400300-{F\	C}	400 × 300 mm	11 × 15 · 25 mm
CT-CHT-P-600400-{F\	C}	600 × 400 mm	12 × 18 · 30 mm
CT-CHT-P-800600-{F\	C}	800 × 600 mm	13 × 18 · 42 mm

Inner corners = (rows − 1) × (cols − 1). Odd × even square counts are used so the pattern has a unique orientation (one black and one white corner square along each edge). Other grids: custom order (CUS) or generate at `calib.tools/tools/generator/` and order as custom.

Specifications

Geometry

Feature position tolerance	±0.3 mm + 0.2 mm/m (any corner to nominal grid, at 20 °C)
Square pitch, mean	±0.05 mm
Orthogonality	±0.02°
Edge radius	R6 mm

Physical

Substrate	Aluminium / LDPE / aluminium composite, 6 mm
Weight	6.6 kg/m²
Pattern	Direct UV print, matte, black on white; optical density ≥ 1.6
Flatness	≤ 0.5 mm per metre, three-point support
Mounting	4 × M4 rear inserts (300 × 200 and larger); ¼″ fixture ACC-A-FIX-14

Environment

Operating temperature	−50 to +80 °C
Thermal expansion	≈ 24 µm/(m·°C)
Outdoor lifetime	up to 3 years

Deliverables

Inspection report CT-IR-<serial> · `.opencv.json`, `.kalibr.yaml`, `.ros2.yaml`, `.halcon.descr`, `.dxf` · CT-UM-GEN manual · Made to order · ships in 2–3 weeks · Options: ILAC-MRA certificate issued by an accredited third-party laboratory, available as SVC-X-CERT-ILAC, -L logo, ACC-A-CSE case.

Compliance and customs

RoHS-compliant materials. HS 9031 80.

Revision history

Rev	Date	Change
A	2026-10	First issue, planning values

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